



Module Presenter's Manual *for* **Programming Embedded Systems**

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1. Introduction

At the end of this course, students will be able to:

- Explain the various data types used in C
- Explain the various types of constants used in C
- Explain how to declare the variables in C
- Explain the various operators use in C
- Explain how to use Arrays, Pointers, Structures, and Enumerated data types in programs
- Explain the use of functions in C
- Explain the common programming errors in C
- Explain the various memory areas supported by the C compiler for the 8051 Microcontroller
- Explain the various memory models supported by the C compiler for the 8051 Microcontroller
- Explain how to access the various memory areas of the 8051 Microcontroller in a C program
- Explain the various data types provided in C for the 8051 Microcontroller
- Explain how to declare and use pointers in the program developed for the 8051 Microcontroller
- Explain how to declare and use functions
- Explain how to work with arrays
- Explain the contents of the file Reg51.h
- Explain how to select the memory model in the C program
- Explain how to use the µVision simulator to develop C program for the 8051 Microcontroller
- Explain how to create an application for the 8051 Microcontroller using C language
- Explain how to debug the C application using the single stepping
- Explain the impact of memory model on the size of the target code
- Explain how to use Proview32 simulator to develop C program for the 8051 Microcontroller
- Explain how to create Intel Hex file in µVision and Proview32 simulators
- Explain the use of arithmetic operators
- Explain the use of unary operators
- Explain how to use bitwise operators to perform masking operations
- Explain how to use the ~ operator to compute Ones and Two's complement of a number
- Explain how to use the left shift and right shift operators
- Explain how to convert ASCII code to packed BCD codes
- Explain how to convert packed BCD codes into ASCII codes
- Explain the architecture of the Cortex M0 processor from ARM
- Explain the programmers model of the Cortex M0 processor
- Explain the register set of the Cortex M0 processor
- Explain the memory organization of Cortex M0 processor
- Explain the interrupt handling and processing of the Cortex M0 processor
- Describe the power control feature in Cortex M0 processor
- Explain the architecture of Atmel SAM D21 processor

2. Information on Session Allocation

Module	Online Sessions (No. of Hours)
Programming Embedded Systems	40

Throughout this Presenter's Manual, the module **Programming Embedded Systems** will be referred to as **PES**.

3. Module Deliverables

To aid the teaching process, following are the deliverables

(Faculties can access eBooks and other components in Onlinevarsity through Aptrack.)

Resources available on Onlinevarsity:

Icons	Feature - Description/Functionality
	Download Book - Student has the option to download the subject related e-book and read offline.
	Glossary - Student can access a list of subject related specialized words with their definitions.
	FAQ - Student can access frequently asked questions and their answers.
	Practice 4 Me - Student can test and evaluate their understanding of module related topics.
	Work Assignments - Student can solve scenario based lab assignments (Hands-on). The faculty will evaluate and give their feedbacks.
	References - Student can access additional subject related material for reading.
	Feedback - Student can provide feedback on the course material.
	Ask to Learn – Student can submit subject related technical queries. Queries submitted will be directed to the particular course coordinator/head.

4. Week-wise Session Schedule

- A Session is of 2 hours duration.

➤ Week-Wise Schedule

Week	Day 1	Day 2	Day 3	Day 4
1	Session 1 PES – TL1	Session 2 PES – TL2	Session 3 PES – TL3	Session 4 PES – TL4
2	Session 5 PES – TL5	Session 6 PES – TL6	Session 7 PES – TL7	Session 8 PES – TL8
3	Session 9 PES – TL9	Session 10 PES – TL10	Session 11 PES – TL11	Session 12 PES – TL12
4	Session 13 PES – TL13	Session 14 PES – TL14	Session 15 PES – TL15	Session 16 PES – TL16
5	Session 17 PES – TL17	Session 18 PES – TL18	Session 19 PES – TL19	Session 20 PES – TL20

PES: Programming Embedded Systems

TL: Online Session

5. Session Coverage

Session No.	Session Title	Session Details	Deliverables' Mapping
1	PES – TL1	All the topics as listed below from Session 1 of <i>Programming Embedded Systems</i> book should be covered in this session. <u>Session 1 – Working with C</u> ➤ Explain the various data types used in C ➤ Explain the various types of constants used in C ➤ Explain how to declare the variables in C ➤ Explain the various operators use in C ➤ Explain how to use Arrays, Pointers, Structures, and Enumerated data types in programs ➤ Explain the use of functions in C ➤ Explain the common programming errors in C	Programming Embedded Systems SG – Session 1 XP – Session 1 TG – Session 1
2	PES – TL2	All the topics as listed below from Session 2 of <i>Programming Embedded Systems</i> book should be covered in this session. <u>Session 2 – 8051 Extension to the C Language</u> ➤ Explain the various memory areas supported by the C compiler for the 8051 Microcontroller ➤ Explain the various memory models supported by the C compiler for the 8051 Microcontroller ➤ Explain how to access the various memory areas of the 8051 Microcontroller in a C program ➤ Explain the various data types provided in C for the 8051 Microcontroller ➤ Explain how to declare and	Programming Embedded Systems SG – Session 2 XP – Session 2 TG – Session 2

Session No.	Session Title	Session Details	Deliverables' Mapping
		use pointers in the program developed for the 8051 Microcontroller ➤ Explain how to declare and use functions ➤ Explain how to work with arrays	
3	PES – TL3	All the topics as listed below from Session 3 of <i>Programming Embedded Systems</i> book should be covered in this session. <u>Session 3 – Developing Embedded Applications Using C</u> ➤ Explain the contents of the file Reg51.h ➤ Explain how to select the memory model in the C program ➤ Explain how to use the µVision simulator to develop C program for the 8051 Microcontroller ➤ Explain how to create an application for the 8051 Microcontroller using C language ➤ Explain how to debug the C application using the single stepping ➤ Explain the impact of memory model on the size of the target code ➤ Explain how to use Proview32 simulator to develop C program for the 8051 Microcontroller ➤ Explain how to create Intel Hex file in µVision and Proview32 simulators	Programming Embedded Systems SG – Session 3 XP – Session 3 TG – Session 3
4	PES – TL4	All the topics as listed below from Session 4 of <i>Programming Embedded Systems</i> book should be covered in this session. <u>Session 4 – Number System and Data Operations</u>	Programming Embedded Systems SG – Session 4 XP – Session 4 TG – Session 4

Session No.	Session Title	Session Details	Deliverables' Mapping
		➤ Explain the use of arithmetic operators ➤ Explain the use of unary operators ➤ Explain how to use bitwise operators to perform masking operations ➤ Explain how to use the ~ operator to compute Ones and Two's complement of a number ➤ Explain how to use the left shift and right shift operators ➤ Explain how to convert ASCII code to packed BCD codes ➤ Explain how to convert packed BCD codes into ASCII codes	
5	PES – TL5	All the topics as listed below from Session 5 of <i>Programming Embedded Systems</i> book should be covered in this session. <u>Session 5 – Control Flow and Functions</u> ➤ Explain how decision making is implemented in C language using if statement ➤ Explain how to use if-else statements to select between two blocks of code ➤ Explain how to use the switch construct ➤ Explain the use of for, while, and do-while loop ➤ Explain how to create infinite loop ➤ Explain how to use functions in C ➤ Explain how to use pass by reference arguments in C ➤ Explain how to create a program to convert a Hexadecimal number to Decimal number	Programming Embedded Systems SG – Session 5 XP – Session 5 TG – Session 5
6	PES – TL6	All the topics as listed below from Session 6 of <i>Programming Embedded Systems</i> book should be covered in this session.	Programming Embedded Systems SG – Session 6

Session No.	Session Title	Session Details	Deliverables' Mapping
		<u>Session 6 – Accessing Memory in C</u> ➤ Explain how to use arrays in the Embedded System applications ➤ Explain the precautions to be taken when using an array of large number of elements ➤ Explain how to use code memory to store constants ➤ Explain how to use pointers in the Embedded System applications ➤ Explain the process of conversion from generic to memory specific pointers and vice-versa ➤ Explain how to use the SFR memory space using C ➤ Explain how to use the bit addressable locations in the internal memory locations ➤ Explain how to store variables at a fixed memory location ➤ Explain how the C compiler uses the internal data memory space to store variables in small mode	XP – Session 6 TG – Session 6
7	PES – TL7	All the topics as listed below from Session 7 of <i>Programming Embedded Systems</i> book should be covered in this session. <u>Session 7 – Programming I/O Ports</u> ➤ Explain the organization and working of the input/output ports of the 8051 ➤ Explain how to read and write data from and to the 8051 port using C programming language ➤ Explain how to interface a simple keyboard to the 8051 microcontroller ➤ Explain how to expand the I/O ports of the 8051 using the 8255	Programming Embedded Systems SG – Session 7 XP – Session 7 TG – Session 7

Session No.	Session Title	Session Details	Deliverables' Mapping
		➤ Programmable Peripheral Interface ➤ Explain how the 8255 PPI communicates with the 8051 ➤ Explain the various modes of operation of the 8255 PPI ➤ Explain the Bit Set/Reset mode of operation of the 8255 PPI ➤ Explain an example of Embedded System which uses 82	
8	PES – TL8	All the topics as listed below from Session 8 of <i>Programming Embedded Systems</i> book should be covered in this session. <u>Session 8 – Data Moment and Airthmetic Operations</u> ➤ Explain the use of timers in the Embedded System applications ➤ Explain the different modes of operations of the timers ➤ Develop C programs to program the timers to operate in different modes ➤ Explain how to calculate the count for generating required time delays ➤ Explain the working of a Real Time Clock ➤ Explain how to configure the DS 12887 RTC using the C program ➤ Explain how to set the time and date in the DS 12887 RTC ➤ Explain how to configure the 8051 timers to operate as counters	Programming Embedded Systems SG – Session 8 XP – Session 8 TG – Session 8
9	PES – TL9	All the topics as listed below from Session 9 of <i>Programming Embedded Systems</i> book should be covered in this session. <u>Session 9 – Programming the 8051 Interrupt</u> ➤ Explain the working of the	Programming Embedded Systems SG – Session 9 XP – Session 9 TG – Session 9

Session No.	Session Title	Session Details	Deliverables' Mapping
		8051 Microcontroller interrupts ➤ Explain how the 8051 processes the interrupts ➤ Explain how to create ISRs using the C program ➤ Explain how to configure the DS 12887 RTC to generate alarms ➤ Explain how to configure the DS 12887 RTC to generate periodic interrupts ➤ Explain how to create watchdog timers using the 8051 timers and DS 12887 RTC ➤ Explain how to expand the number of external interrupts in an 8051 based system	
10	PES – TL10	All the topics as listed below from Session 10 of <i>Programming Embedded Systems</i> book should be covered in this session. <u>Session 10 – Serial Communication Using the 8051</u> ➤ Explain the working of serial communication ➤ Explain the conversion of parallel data to serial data and vice versa ➤ Explain the asynchronous mode of serial communication ➤ Explain the RS 232 standard for serial communication ➤ Explain how 8051 performs serial communication ➤ Explain how to set the baud rate for serial communication in the 8051 ➤ Explain how to use serial interrupt to manage serial communication	Programming Embedded Systems SG – Session 10 XP – Session 10 TG – Session 10
11	PES – TL11	All the topics as listed below from Session 9 of <i>Programming Embedded Systems</i> book should be covered in this session. <u>Session 11 – Programming</u>	Programming Embedded Systems SG – Session 11 XP – Session 11 TG – Session 11

Session No.	Session Title	Session Details	Deliverables' Mapping
		<u>ADC and DAC Interfaces</u> ➤ Explain the use of Analog to Digital Converters and Digital to Analog Converters in Embedded Systems ➤ Explain the different types of Analog to Digital Converters ➤ Explain how to read data from parallel Analog to Digital Converters ➤ Explain how to read data from serial Analog to Digital Converters ➤ Explain how to write data to Digital to Analog Converters ➤ Explain how to use sensors to read the physical environment data	
12	PES – TL12	All the topics as listed below from Session 10 of <i>Programming Embedded Systems</i> book should be covered in this session <u>Session 12 – ARM Microcontrollers</u> ➤ Explain the architecture of the Cortex M0 processor from ARM ➤ Explain the programmers model of the Cortex M0 processor ➤ Explain the register set of the Cortex M0 processor ➤ Explain the memory organization of Cortex M0 processor ➤ Explain the interrupt handling and processing of the Cortex M0 processor ➤ Describe the power control feature in Cortex M0 processor ➤ Explain the architecture of Atmel SAM D21 processor	Programming Embedded Systems Session 12 Session 12 Session 12
13	PES – TL13	All the topics as listed below from Lab Session 1 of <i>Programming Embedded Systems</i> book should be covered in this session	Programming Embedded Systems (LAB) Session 1

Session No.	Session Title	Session Details	Deliverables' Mapping
		<u>Session 1 – Developing Embedded Application Using C</u> ➤ Use µVision Simulator to create, compile, and execute a C program for the 8051 microcontroller	
14	PES – TL14	All the topics as listed below from Lab Session 2 of <i>Programming Embedded Systems</i> book should be covered in this session <u>Session 2 – Airthmetic and Logic Operations</u> ➤ Create a program to add two numbers ➤ Create a program to subtract two numbers ➤ Create a program to compute remainder of division of two numbers ➤ Create a program to perform bitwise OR operation ➤ Create a program to perform bitwise AND operation ➤ Create a program to right shift a number ➤ Create a program to convert a packed BCD number into equivalent ASCII digits	Programming Embedded Systems (LAB) Session 2
15	PES – TL15	All the topics as listed below from Lab Session 3 of <i>Programming Embedded Systems</i> book should be covered in this session <u>Session 3 – Control Flow and Function</u> ➤ Create a program to find larger of two numbers ➤ Create a program to find factorial of a number using a function ➤ Create a program to convert a Hexadecimal number to its equivalent Decimal form and displays individual digits in ASCII code	Programming Embedded Systems (LAB) Session 3

Session No.	Session Title	Session Details	Deliverables' Mapping
16	PES – TL16	All the topics as listed below from Lab Session 4 of <i>Programming Embedded Systems</i> book should be covered in this session <u>Session 4 – Accessing Memory in C</u> ➤ Create a program to sort the numbers stored in an array in descending order ➤ Create a program to sort the numbers stored in an array in external memory location ➤ Create a program to demonstrate the effect of array overrun in the internal data memory	Programming Embedded Systems (LAB) Session 4
17	PES – TL17	All the topics as listed below from Lab Session 5 of <i>Programming Embedded Systems</i> book should be covered in this session <u>Session 5 – Programming the I/O ports</u> ➤ Create a program to read data from Port 0 bit 0 and write the complement on Port 0 bit 0 ➤ Create a program to output a squarewave with 66% duty cycle on pin 0 of Port 0 ➤ Create a program to configure the 8255 PPI to operate in Mode 0	Programming Embedded Systems (LAB) Session 5
18	PES – TL18	All the topics as listed below from Lab Session 6 of <i>Programming Embedded Systems</i> book should be covered in this session <u>Session 6 – Programming the 8051 Timers</u> ➤ Create a program to configure the Timer 1 in Mode 0 to generate a delay of 500μs ➤ Create a program to configure the Timer 1 in Mode 2 to generate a delay of 100μs ➤ Create a program to read the	Programming Embedded Systems (LAB) Session 6

Session No.	Session Title	Session Details	Deliverables' Mapping
		time of the DS 12887 RTC	
19	PES – TL19	All the topics as listed below from Lab Session 7 of <i>Programming Embedded Systems</i> book should be covered in this session <u>Session 7 – Programming the 8051 Interrupts</u> ➤ Create a program to handle the Timer 1 interrupt ➤ Create a program to implement a watchdog timer using Timer 1	Programming Embedded Systems (LAB) Session 7
20	PES – TL20	All the topics as listed below from Lab Session 8 and 9 of <i>Programming Embedded Systems</i> book should be covered in this session <u>Session 8 – Serial Communication Using the 8051</u> ➤ Create a program to generate serial output on pin 0 of Port 1 ➤ Create a program to transmit a byte of data serially using MOD 0 of UART <u>Session 9 – Programming ADC and DAC Interfaces</u> ➤ Create a program to read data from an Analog to Digital Converter 0804 ➤ Create a program to read data from 0808ADC ➤ Create a program to output sine wave using 0808 DAC	Programming Embedded Systems (LAB) Session 8 and 9

6. Library References

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| ➤ Embedded Microcomputer Systems by Jonathan W. Valvano |
| ➤ Embedded Systems: Architecture, Programming and Design by Raj Kamal |

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